BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

LESSON 019 · CPU Performance

Name: _____ Class: _____ Date: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

Do Now · Recall previous learning

Q1. What is the raw bitmap size in bits before metadata or compression?

Q2. For cost = price × quantity, which is an input?
A. Calculated total cost
B. Printed receipt
C. Quantity
D. Displayed bill

Q3. Why use pseudocode when designing an algorithm?

Q4. A flowchart must accept ages 13–18 and reject others. “A diamond only shows calculations with no conditions.” Correct this. How should it check age?

BLACK PEN Answer all questions first. GREEN PEN Then: tick and rewrite mistakes.

I Do · Read new knowledge

Key vocabulary

Clock speed — The number of clock cycles the CPU carries out per second, measured in hertz.

Hertz (Hz) — Unit of clock speed: one clock cycle per second.

Cache — Small, fast memory that stores frequently used data and instructions for the CPU.

CPU core — Part of a CPU that processes instructions.

Parallel processing — Processing more than one task at the same time using multiple CPU cores.

New learning

Clock speed: Clock speed is measured in hertz (Hz). A higher clock speed means the CPU carries out more clock cycles each second, so it can generally process instructions faster.

Cache size: Cache is small, fast memory used to store frequently used data and instructions. A larger cache can reduce the need to fetch them from slower main memory (RAM), improving performance.

Number of cores: A core processes instructions. A processor with more cores can process more tasks at the same time, which can improve performance when several programs are running.

Compare before you decide: Look at all three factors: clock speed, cache size and number of cores. For example, one CPU may have a higher clock speed while another has more cores and a larger cache. Explain which factors matter for the stated task.

Worked example

Using the CPU A and CPU B specifications shown, explain TWO reasons CPU B may be better for running several programs at the same time. [4 marks]

1. Read the specifications shown: CPU A has the higher clock speed. CPU B has more cores and a larger cache. The task involves running several programs.

2. First reason: CPU B has four cores rather than two [1]. More cores can process more tasks at the same time, improving multitasking [1].

3. Second reason: CPU B has 8 MB cache rather than 4 MB [1]. More frequently used data and instructions can be accessed quickly, reducing slower RAM access [1].

4. Check the command word 'explain': each reason must link a CPU characteristic to an improvement. CPU A's higher clock speed is a valid comparison, but we cannot guarantee which CPU is faster in every task.

We Do · Practise with guidance

Guided 1

Compare CPU C and CPU D

Worked method: CPU C: 3.6 GHz, 2 cores, 4 MB cache. CPU D: 3.0 GHz, 6 cores, 12 MB cache. A pupil runs several programs at the same time. Explain TWO reasons why CPU D may be a better choice. [4 marks]

Guided 2

Clock speed

Worked method: CPU E runs at 3 GHz and CPU F at 2 GHz. Identify which has the higher clock speed and explain one possible performance advantage. [2 marks]

Guided 3

Cache size

Worked method: A processor is upgraded from 2 MB of cache to 8 MB. Explain how this could improve CPU performance. [2 marks]

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

Code First · Predict → Write → Test → Improve

Predict the message printed.

```
clockA = 3
clockB = 2
if clockA > clockB then
    print("A has higher clock speed")
endif
```

predict

Question 1

State the three characteristics of a CPU that affect performance. [3 marks] Source: Original J277-style question based on OCR J277 1.1.2.

Question 2

A college chooses a CPU for computers running several applications. CPU G: 3.9 GHz, 2 cores, 6 MB cache. CPU H: 3.1 GHz, 8 cores, 16 MB cache. Explain TWO reasons why CPU H could be better for this use. [4 marks] Source: Original J277-style application question based on OCR J277 1.1.2.

Question 3

Explain how clock speed, cache size and the number of cores each affect CPU performance. [6 marks] Source: Adapted from your Paper 1 revision booklet, section 1.1.2.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

Exit Ticket · Answer before checking

A processor with more cores can improve performance when several programs are running.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

Page 1 onwards · Do Now

Q1. Width × height × bits per pixel.

Q2. C — Quantity

Q3. Pseudocode shows steps in order without needing one programming language's exact syntax.

Q4. A diamond is a decision. Its branches follow different conditions. Check that each input follows the right path.

I Do / We Do pages · Guided practice

Q1. CPU C: 3.6 GHz, 2 cores, 4 MB cache. CPU D: 3.0 GHz, 6 cores, 12 MB cache. A pupil runs several programs at the same time. Explain TWO reasons why CPU D may be a better choice. [4 marks]

Q2. CPU E runs at 3 GHz and CPU F at 2 GHz. Identify which has the higher clock speed and explain one possible performance advantage. [2 marks]

Q3. A processor is upgraded from 2 MB of cache to 8 MB. Explain how this could improve CPU performance. [2 marks]

You Do pages · Code First

```
clockA = 3
clockB = 2
if clockA > clockB then
    print("A has higher clock speed")
endif
```

Expected result: A has higher clock speed

You Do pages · Independent questions

Q1. Clock speed [1]; cache size [1]; number of CPU cores [1].

Q2. CPU H has eight cores rather than two [1], so it can process more tasks at the same time [1]. CPU H has 16 MB cache rather than 6 MB [1], so more frequently used instructions/data can be accessed quickly, reducing slower RAM accesses [1]. CPU G has the higher clock speed; the best choice depends on the workload.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

Original practice marking points, not an official OCR mark scheme.

Q3. Clock speed: a higher clock speed provides more clock cycles each second [1], allowing the CPU to process instructions faster [1]. Cache size: a larger cache can store more frequently used data/instructions [1], reducing access to slower main memory [1]. Number of cores: more cores can process more tasks at the same time [1], improving multitasking/performance for tasks that can run in parallel [1]. This point-by-point allocation is a teaching model; assess extended answers against the relevant question criteria.

Exit Ticket · Model answer

True

Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

1. Copy the whole original question:

2. Write a complete new answer in GREEN PEN. Show working if needed: